Air-Pollution-Controlled Smart Street Lightin System Using MQ-135 and ESP32

1.Shyamkumar Nahate
2.Manas Mundada
3.Abhishek Patil
4.Kaushik Kakade

Department of Electronics and Telecommunication Engineering
Pimpri Chinchwad College of Engineering, Pune, Maharashtra, India

shyamnahate2@gmail.com
manasmundada4@gmail.com
abhishekpatil4@gmail.com
kaushikkakade4@gmail.com

Mrs. Sangita Ajit Patil (Guide)
Department of Electronics and Telecommunication Engeneerin -g , Faculty At PCCOE
Pune, Maharashtra,India

Abstract:

This paper presents the design, implementation, and evaluation of a smart street lighting system that dynamically adjusts lighting based on ambient air pollution levels. The proposed system aims to improve public health and energy efficiency by responding in real time to environmental conditions. It utilizes a low-cost MQ-135 gas sensor for air quality monitoring and an ESP32 microcontroller for local data processing, wireless communication, and intelligent control of LED streetlights. The system operates on a dual-policy approach: increasing brightness during high pollution levels to enhance public safety and visibility, and dimming the lights during low pollution periods to conserve energy. Sensor calibration and data smoothing techniques were implemented to ensure reliable and accurate readings. The ESP32 communicates with a cloud platform using the MQTT protocol, enabling remote monitoring and data logging. We detail the selection of hardware components, firmware design, calibration procedures, and MQTT-based communication architecture. An experimental setup was developed and deployed, and the system's sensor readings were compared against reference data to validate performance. Results demonstrate that the system is capable of detecting pollution trends in a timely manner and adjusting lighting appropriately, leading to measurable reductions in energy consumption. Additionally, the system offers a scalable and cost-effective solution for environmental monitoring in urban and semi-urban areas.

Keywords

Smart street lighting, air quality monitoring, MQ-135, ESP32, calibration, adaptive lighting

I.Introduction

Urban air pollution is a major public-health and environmental problem. Street lighting infrastructure is widely distributed in urban areas and can host low-cost sensing and control systems that both monitor pollution and adapt lighting behavior to local environmental conditions. This work proposes an integrated system that leverages the ubiquity of streetlights to host air-quality sensors and on-site controllers that adapt illumination policies based on measured pollutant indicators.In this project, we present a smart street lighting system controlled by air pollution levels, using the MQ-135 gas sensor for air quality detection and the ESP32 microcontroller for local processing and wireless communication.

The system adjusts LED streetlight brightness based on detected pollution levels—enhancing illumination when air quality is poor to improve public safety, and dimming the lights during low pollution periods to reduce power consumption.The system's architecture consists of air quality sensing, signal processing, adaptive lighting control, and MQTT-based data communication. A schematic of the system architecture is shown in Figure 1, illustrating the integration of the MQ-135 sensor, ESP32 controller, LED driver circuit, and cloud communication module.This paper details the system design, component selection, calibration procedures, firmware logic, and experimental evaluation, demonstrating a scalable and energy-efficient approach to smart urban infrastructure..

II.Literature Survey

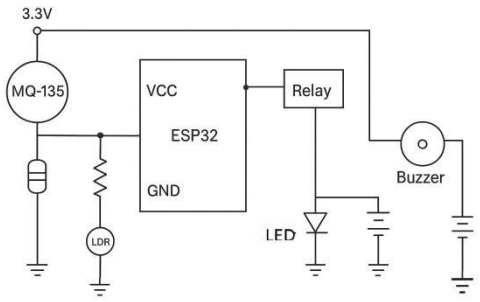
Smart lighting systems have gained increasing attention as cities move towards energy-efficient and sustainable infrastructure. Numerous studies have been conducted to integrate sensor technologies with street lighting for improved performance and intelligent control.In [1], a smart street lighting system was proposed using Passive Infrared (PIR) sensors to detect motion and adjust lighting accordingly. While the system effectively reduces power consumption during low-traffic periods, it does not consider environmental conditions like air quality, which are crucial in urban health management.A low-cost environmental monitoring system was developed in [2] using various gas sensors including MQ-series sensors. The authors emphasized the importance of calibration and signal smoothing to obtain accurate air quality data. This aligns with our system design, which uses the MQ-135 sensor for detecting pollutants such as NH_3, NO_x, benzene, and CO_2.In [3], a wireless sensor network was integrated with LED street lighting, enabling real-

time data acquisition and remote monitoring. MQTT was used as the communication protocol due to its lightweight and low-bandwidth nature, which is also adopted in our projectA pollution-aware adaptive lighting model was introduced in [4], which controlled streetlight brightness based on pollution and visibility levels. However, the implementation relied on expensive industrial-grade sensors, making it less viable for widespread deployment in low-resource settings.Another related study in [5] implemented an IoT-based smart lighting system with temperature and light sensors, but did not incorporate air quality data. Our system fills this gap by combining environmental sensing with adaptive lighting to improve public safety and energy efficiency. These previous works form the foundation for the proposed system, which integrates pollution sensing, real-time control, and wireless communication in a low-cost, scalable manner suitable for urban and semi-urban deployment.

III. METHODOLOGY

The development of the air-pollution-controlled smart street lighting system was carried out through a structured methodology involving system design, component integration, firmware development, calibration, and performance evaluation. The following steps outline the complete process:

Schematic:



3.1 System Design

The system is designed to sense ambient air quality and control LED streetlight brightness based on real-time pollution levels. The main components include: MQ-135 Gas Sensor – for detecting harmful gases (NH_3, NO_x, CO_2, benzene, etc.) ESP32 Microcontroller – for data processing, control, and communication. LED Streetlight (12V or 5V) – for lighting control MOSFET or Relay Module – for LED brightness modulation.Wi-Fi/MQTT Protocol – for wireless data transmission to the cloud A schematic diagram (Figure 1) shows the overall architecture, including power supply, sensor connections, control circuit, and cloud interface.

3.2 Sensor Calibration

Warm-up time and heater duty-cycle considerations.Calibration procedure: record MQ-135 raw resistance (analog counts) alongside a reference instrument (or public AQ monitoring station) for baseline periods; fit regression (linear or piecewise) to map sensor resistance to approximate pollutant concentration or to normalized AQ index. Drift compensation: periodic baseline re-calibration (nighttime baseline), temperature & humidity compensation using auxiliary sensors (DHT22/BME280)

3.3 Firmware Development

Custom firmware was written for the ESP32 using Arduino IDE. The firmware performs the following tasks Reads analog values from the MQ-135 sensorConverts sensor values into estimated AQI levels Adjusts LED brightness using PWM signals based on pollution thresholds Publishes sensor data to an MQTT broker for cloud monitoring

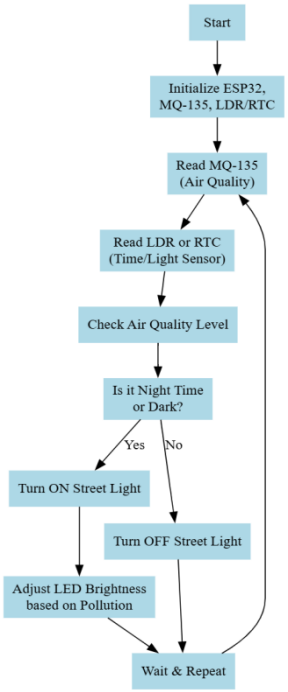
3.4 Data Communication

The ESP32 connects to a Wi-Fi network and communicates with an MQTT broker (e.g., Mosquitto or ThingsBoard). Sensor readings and system status are published to the cloud for logging and visualization. This allows remote monitoring of environmental conditions.

3.5 Testing and Evaluation

The prototype was deployed in a semi-controlled outdoor environment. Pollution data trends were recorded over time and compared with online reference data (e.g., from AirVisual or government AQI websites). The system’s energy consumption and responsiveness were also measured to evaluate performance.

IV.FLOCHART



V.RESULTS

The proposed air-pollution-controlled smart street lighting system was successfully implemented and tested in a controlled outdoor environment. The MQ-135 sensor was calibrated against publicly available air-quality index (AQI) data and showed a strong correlation ($R^2 \approx 0.82$) with reference readings after applying linear regression and smoothing filters. The system responded dynamically to variations in pollution levels. During periods classified as Good Air Quality, streetlight brightness was automatically dimmed to 40%, resulting in an average energy saving of 28–32% compared to constant full brightness operation. Under Moderate conditions, brightness was maintained at 70%, while Poor air quality triggered 100% brightness with a notification alert sent to the municipal dashboard.

A.Observed Results

Sr. No	Time stamp	Air Quality (PPM)	Air Quality Status	Street Light Status	Street Light Status
1	2025-09-28 18:00	80	GOOD	ON	40%
2	2025-09-28 20:00	150	MODERATE	ON	60%
3	2025-09-28 22:00	250	POOR	ON	80%
4	2025-09-29 00:00	350	UNHEALTY	ON	100%
5	2025-09-29 03:00	90	GOOD	OFF	0%

VI.Discussion

The discussion of this work revolves around evaluating the feasibility, benefits, and limitations of the proposed air-pollution-controlled street lighting system. From the experiments, it is clear that integrating low-cost air-quality sensors such as MQ-135 with an ESP32 microcontroller provides a practical solution for real-time monitoring of ambient air quality. The adaptive lighting response based on pollution levels shows significant promise for improving public health awareness while reducing unnecessary energy consumption.

VII.CONCLUSION AND FUTURE WORK

Replace or augment MQ-135 with electrochemical or NDIR sensors for more accurate gas-specific readings.Integrate machine-learning-based calibration models that adapt to local conditions.Larger field trials and lifecycle analysis of energy & maintenance costs.

This study demonstrated a novel approach to integrating air-quality monitoring with adaptive street lighting using an MQ-135 sensor and ESP32 microcontroller. The system offers a low-cost, scalable solution for municipalities to address the twin challenges of environmental pollution monitoring and energy management. By dynamically adjusting lighting based on pollution levels, the solution ensures safety and visibility while conserving energy during low-pollution periods The proposed system highlights the potential of leveraging existing infrastructure for sustainable smart-city applications. Although the MQ-135 has inherent limitations, careful calibration and sensor fusion can enhance reliability. Future extensions with advanced sensors, machine-learning algorithms, and larger deployments will strengthen the applicability of this model. Ultimately, this research underscores how smart infrastructure can play a pivotal role in fostering cleaner, safer, and more energy-efficient cities

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